

Parallel Computing Previous Year QNs

(Year – 2074, 2075, 2076, 2077, 2079, 2080, 2081)

Short Questions:

2074 Short QNs.

1. What is parallel computing? Who is using parallel computing?
2. Explain Gustafson-Barsis's law with a suitable example and compare it with Amdahl's law.
3. Why are message passing models used? What is included and not included in the MPI standard?
4. What is the shared memory model? Explain its uses.
5. What is data parallelism? Explain machine model and CU/PU overlap.
6. How can you solve a linear system? Use Jacobi iteration method to solve the following equations:

$$6x_1 - 2x_2 + x_3 = 11$$

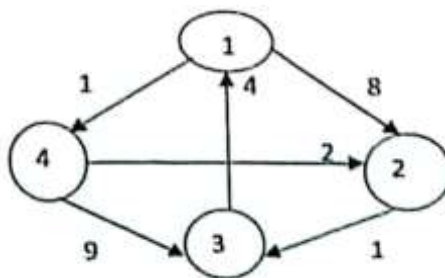
$$x_1 + 2x_2 - 5x_3 = -1$$

$$-2x_1 + 7x_2 + 2x_3 = 5$$

7. Explain divide and conquer technique. Write down the features of hypercube.
8. Write short notes on:
 - a. Iso-efficiency Metric
 - b. Monte Carlo Methods
 - c. Clustering

9. How does OpenMP provide shared memory programming environment? What is the memory consistency model supported by OpenMp?
10. What is Routing Mechanism? Differentiate between deterministic routing and adaptive routing.
11. What is pipelining? Describe the speed up gain due to pipelining.
12. What do you mean by parallel reduction operation? Differentiate between single reduction and multiple reduction operations on a single set of data.
13. What is hypercube connection? Explain forward and reverse phase of prefix computation on hypercube.
14. Define Amdahl's law. What do you conclude from Amdahl's law?
15. Write short notes on:
 - a. MIMD
 - b. Task Allocation
 - c. Monte Carlo Method
16. Write Floyd-Warshall algorithm. Use this algorithm to find shortest path between all pairs of vertices of figure given below:

2075 Short QNs.



17. What is NUMA memory architecture? Differentiate between one-to-all broadcast and all-to-one reduction.
18. Explain Gustafson-Barsis's law and compare it with Amdahl's law.

2076 Short QNs.

19. What is PRAM Model? What are the subclasses of PRAM?
20. Differentiate between parallel and serial computing. Explain Foster's design methodology.
21. Explain sieve of Eratosthenes algorithm. Use this algorithm to find all the prime numbers between 1 to 20.
22. What is parallel reduction operation? Explain its types.
23. Write Floyd-warshall algorithm. Trace it with suitable example.
24. What is data parallelism? Explain machine model and CU/PU overlap.

25. What are the criteria that are used to evaluate the cost and performance of static interconnection networks?
26. Explain the features of parallel object-oriented programming.
27. Explain Amdahl's law with suitable example. Why actual speed ups are always less in Amdahl's law?
28. Explain distributed shared memory system. Write its advantages.
29. What are parallel reduction algorithms? How they are used in PRAM architecture.
30. Define message passing. Use Eratosthenes algorithm to find all the prime number between 1 to 20.
31. Discuss different parallel approaches including data parallelism and control parallelism.
32. Write short notes on:
 - a. SIMD
 - b. Pipelining
 - c. Canon's algorithm

2077 Short QNs.

33. Define Pipelining.
34. What is NUMA memory architecture.
35. What is the memory consistency model supported by OpenMP?
36. Explain Iso-efficiency matrix.
37. Write down the application of Monte-Carlo method.
38. Explain cannon's algorithm.
39. What are the steps of Hyper Quicksort?
40. Differentiate between deterministic routing and adaptive routing.

2079 very Short QNs.

41. Define Amdahl's law. Write down the limitation of Amdahl's law.
42. Explain the steps of Foster's design methodology.
43. Define message passing model. What are the advantages and disadvantages of message passing model?
44. Differentiate between data and task parallelism.
45. Explain any one example of combinatorial search in detail.
46. Explain Karp-Flatt metric with suitable example.

2079 Short QNs.

47. How will you define speed up if a number of processors is added to perform fraction of work in parallel?
48. How does OpenMP provide a shared memory programming environment?
49. What are the classifications of routing mechanism?
50. Differentiate between data and task parallelism.
51. Define the term agglomeration.
52. Explain Monte Carlo method.
53. What is NUMA memory architecture?
54. Explain about loop jamming.

2080 very Short QNs.

55. Explain the concept of speed up by applying Amdahl's law. What is speed up if no part of the code can be parallelized? What is the speed up if 50% of the code can be parallelized?
56. Why do we use MPI? Discuss basic concept of the MPI.
57. Write Floyd Warshall algorithm. Trace it with suitable example.
58. Explain Hyper quicksort algorithm with example.
59. Explain in detail about Flynn's classification.
60. How can you solve linear system? Use jacobi iteration method to solve the following equations.

$$\begin{aligned} 6x_1 - 2x_2 + x_3 &= 11 \\ x_1 + 2x_2 - 5x_3 &= -1 \\ -2x_1 + 7x_2 + 2x_3 &= 5 \end{aligned}$$

2080 Short QNs.

61. Explain divide and conquer technique.
62. Differentiate between data and task parallelism.
63. Define parallel programming.
64. Explain Conjugate gradient method.
65. Define Karp-Flatt Metric.
66. What is OpenMP?
67. Explain about Loop Unrolling.
68. Define hyper quicksort.

2081 very Short QNs.

69. Explain Amdahl's law for measuring speedup performance with the help of an example.
70. Explain Monte Carlo methods. Write down its application.
71. What is shared memory model? Write its advantages and disadvantages.
72. What is message passing model? Write down its advantages and disadvantages.
73. Use Eratosthenes algorithm to find all the prime numbers between 1 to 30.
74. Write short notes on (any two):
- Isoefficiency Metric
 - Gustafson-Barsis Law
 - Fast Fourier Transform (FFT) algorithms

2081 very Short QNs.

Long Questions:

1. Explain Canon's algorithm. Use Eratosthenes algorithm to find all the prime numbers between 1 to 40.
2. What do you mean by combinatorial search? Explain any one example in detail.
3. Explain Karp-Flatt metric. Find the value of 'e'. Consider the following speedups obtained by a certain parallel application:

No. of Processors	2 CPUs	3 CPUs	4 CPUs	5 CPUs	6 CPUs	7 CPUs	8 CPUs
Ψ	1.73	2.81	3.11	3.57	4.00	4.38	4.82

2074
Long QNs.

4. What is pipelining? Describe the speedup gain due to pipelining.
5. Explain hypercube algorithm. Write down the features of hypercube. Explain forward and reverse phase of prefix computation on hypercube.
6. Explain Karp-Flatt metric. Find the value of 'e'. Consider the following speedups obtained by a certain parallel application:

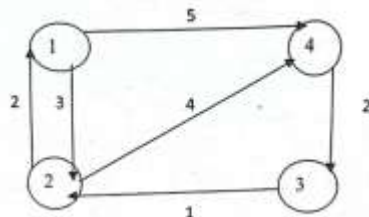
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Ψ	1.73	2.81	3.11	3.57	4.00	4.38	4.82

2076
Long QNs.

7. Explain Karp-Flatt metric. Find the value of 'e'. Consider the following speedups obtained by a certain parallel application:

No. of Processors	2 CPUs	3 CPUs	4 CPUs	5 CPUs	6 CPUs	7 CPUs	8 CPUs
Ψ	1.73	2.81	3.11	3.57	4.00	4.38	4.82

8. What do you mean by combinatorial search? Explain alpha-beta pruning in detail.
9. Explain Floyd-warshall algorithm. Use this algorithm to find the shortest path between all pairs of vertices of figure given below.



2077
Long QNs.

10. What is parallel computing? Explain different paradigms of parallel computing. Also discuss merits and demerits of parallel computation.
11. Explain the architecture of distributed and shared memory computer. What are the major differences between distributed and shared memory computers?
12. Explain Flynn's classification of computer architectures with its block diagram.
13. Explain Floyd-Warshall algorithm with suitable example.

2079
Long QNs.

14. Explain Karp-Flatt metric. Find the value of 'e'. Consider the following speedups obtained by a certain parallel application:

No. of Processors	2 CPUs	3 CPUs	4 CPUs	5 CPUs	6 CPUs	7 CPUs	8 CPUs
Ψ	1.17	2.33	3.23	3.53	4.26	4.56	4.77

2080
Long QNs.

15. Explain Canon's algorithm. Use Eratosthenes algorithm to find all the prime numbers between 1 to 40.
16. Explain the architecture of distributed and shared memory computer. What are the major differences between distributed and shared memory computers?
17. Define clustering. What are the differences between parallel computing, distributed computing and cloud computing?

18. What do you mean by combinatorial search? Explain any one example in detail.

19. Define Foster's design methodology. Explain its steps in detail.

20. What is pipeline and how it works? Explain about Flynn's taxonomy.

21. Write Floyd-Warshall algorithm. Trace it with suitable example.

2081
Long QNs.

Advanced Database Design Previous Year QNs

(Year – 2074, 2075, 2076, 2077, 2079, 2080, 2081)

Short Questions:

1. Explain four desirable properties of a transaction. Describe transaction state diagram in brief. [ACID Properties]
2. What is lock? Explain shared and exclusive locks with example.
3. Explain Sparse, Dense and Multi-level index with example.
4. What is database recovery? Explain shadow paging with example.
5. Explain the importance of data integrity in DBMS.
6. What is a deadlock? How are deadlocks prevented?
7. List out factors for measuring performance of disk. Explain hashing in brief.
8. List out factors for measuring cost of query. Describe transformation rule with example.

2074 Short QNs.

9. What is DBMS? List out the functions of DBMS in brief.
10. Explain RAID technology with example.
11. Describe the importance of concurrency control and explain any one method in brief.
12. Explain primary and secondary index with example.
13. What is deadlock? What are the necessary conditions for deadlock?
14. Describe one method for deadlock recovery.
15. What is query optimization? Describe query optimization process in brief.
16. Write short notes on:
 - a. a. Measures of Query Cost
 - b. b. Fixed length records and variable length records.

2075 Short QNs.

17. What are the advantages of DBMS over file system? Describe in brief.
18. Explain join dependency with example.
19. How can you convert an ER diagram into table? Discuss with example.
20. Explain Sparse, Dense and Multi-level index with example.
21. Explain RAID technology up to three levels with example.
22. Explain state diagram of transaction process in brief.
23. Explain the data integrity in brief.
24. Write short notes (any two):
 - a. Data warehouse
 - b. 2PL
 - c. Shadow paging

2076 Short QNs.

25. Explain join dependency with example.
26. Mention the desirable properties of transaction. Also describe state diagram of transaction.
27. Explain sparse and dense index with example.
28. What are the advantages of DBMS over file system?
29. What is deadlock? How are deadlocks prevented?
30. Differentiate between physical and logical data independence.
31. How do you increase the performance of database tuning?
32. Why do we need RAID? Explain RAID technology up to two levels with example.

2077 Short QNs.

33. Define transaction.
34. List out any three differences between DBMS and RDBMS.
35. Define memory hierarchy.
36. Why do we use outer join?
37. Mention the role of indexing.
38. What are the characteristics of memory?
39. How do you recover data from loss?
40. Is DBMS useful for research?

2079 very
Short QNs.

41. What is two-phase locking? Explain with example.
42. Mention the factors to evaluate indexing. Explain primary and ordered indexing with example.
43. What is query optimization? Write down its importance in brief.
44. Define ACID properties of a transaction with regard to DBMS.
45. Explain various types of failures in DBMS.
46. Define RAID. Explain this technology up to three levels.

2079
Short QNs.

47. What is schema of a database?
48. Define foreign key.
49. What is hashing?
50. Write the statement of second normal form.
51. What are the performance measures of disks?
52. What are the steps in query decomposition?
53. What is multi-level index?
54. What is inheritance? How is it useful?

2080
Short QNs.

55. What is check point in database recovery? How does it help in database recovery?
56. Explain the life cycle of a transaction with state diagram.
57. What is indexing? What are the factors for evaluating indices? Explain primary indexing.
58. Describe join dependency and fifth normal form (5NF) with example.
59. Explain deadlock and starvation. Discuss time stamp-based protocol for concurrency control.
60. What are the advantages of using DBMS over file processing system?

2080 Short QNs.

61. How do you measure performance of disks?
62. What are the functions of Buffer manager?
63. How does B-tree differ from a B+ tree?
64. What are the factors for evaluating indices?
65. What are two differences between DBMS and RDBMS?
66. Define Primary and Candidate key with examples.
67. Define hash function.
68. What is normalization?

2081 very
Short QNs.

69. What are the concurrency control problems that may occur in database? Describe two-phase locking with example.
70. How does database failure occur? How do you recover database?
71. How does bucket overflow occur and how do you handle it?
72. What is indexing? Explain primary indexing with diagram and example.
73. How does DBMS overcome the problems of file processing system?
74. Write short notes on (any two):

2081
Short QNs.

- a. Query Optimization
- b. Fixed-length record and variable-length record
- c. Shadow paging

Long Questions:

1. What is query processing? Explain steps in query processing with diagram and example.
2. Compare various file organizations with example. Also mention storage device hierarchy in brief.
3. Explain RAID technology up to three levels with example.

2074 Long QNs.

4. What is query processing? Explain steps in query processing with diagram.
5. What is file organization? Explain various file organization with their advantages and disadvantages.
6. Explain:
 - a. What are the various types of failures that occur in database? Discuss the log-based recovery mechanism.
 - b. Describe the properties of transactions in brief.

2075 Long QNs.

7. What is time stamping? How can it be used for concurrency control? Describe with the help of example.
8. What is Query processing? Describe query processing mechanism in detail.
9. List out factors for measuring performance of disk. Compare various file organizations with example. Also mention storage device hierarchy in brief.

2076 Long QNs.

10. Describe the importance of data integrity in DBMS.
11. What is query processing? Explain steps in query processing with diagram and example.
12. Differentiate between sequential and random file organization. Explain storage hierarchy in brief.

2077 Long QNs.

13. What is file organization? Explain various file organizations with their merits and demerits.
14. Describe the query processing in detail.
15. What are the advantages of DBMS over file system? Explain in brief.
16. Write short notes on:
 - a. Time stamp ordering protocol
 - b. Deadlock prevention

2079 Long QNs.

17. What is query processing? Explain steps involved in query processing with example.
18. Explain RAID (Redundant Array of Independent Disks). Discuss pros and cons of RAID 0 and RAID 1 Level.
19. Discuss various types of file organization (any three) with their advantages in brief.
20. Write short notes on (any two):
 - a. Fixed length and variable length records
 - b. Memory Hierarchy
 - c. Data independence

2080 Long QNs.

21. Discuss the properties of transaction in DBMS. Explain the states through which a transaction passes during execution.

2081
Long QNs.

22. Explain RAID technology up to (0 and 1) level in database. Also mention its importance.

23. Why is lossless-join important? Explain fifth normal form (5NF) with examples.

24. What is Query processing? What are the steps involved in query processing? Explain with examples.

Cloud Computing Previous Year QNs

(Year – 2076, 2077, 2079, 2080, 2081)

Short Questions:

1. What do you mean by cloud? Describe the basic characteristics of cloud.
2. Explain Platform as a Service (PaaS) and its characteristics.
3. Define virtualization. What is the role of virtualization in cloud computing?
4. Why is cloud migration needed? Explain the model of migration into cloud.
5. Explain the Service-Oriented Architecture (SOA) and its characteristics.
6. Explain the service provided by Amazon EC2 services from a user perspective.
7. Explain the cloud security challenges.
8. Write short notes on:
 - a. Grid computing
 - b. Risk assessment in cloud

2076 Short QNs.

9. Explain in short different participants of cloud.
10. What are benefits of using cloud models?
11. Explain advantages of using CaaS model.
12. Explain characteristics of Amazon EC2 service.
13. What are different building blocks of data centers?
14. Explain Service-Oriented Architectures (SOA) with its characteristics.
15. Explain in short cloud security architecture design.
16. Explain different approaches for host security.

2077 Short QNs.

17. Write short note on IaaS model.
18. Write in short on Public Cloud Computing Deployment Models.
19. What are different advantages of CaaS?
20. What are key characteristics of SaaS software?
21. Explain in short single-purpose architectures to multi-purpose architectures.
22. What are different approaches used for host security.
23. What are different factors involved in disaster recovery planning.
24. Explain in short Distributed Computing in Grid and Cloud.

2079 very Short QNs.

25. Explain the concept of implementing network intrusion detection in the cloud.
26. What are the different approaches for host security in cloud computing?
27. What are the benefits of data center virtualization?
28. What are the principal characteristics of SOA?
29. Explain CaaS with its advantages in short.
30. Explain Amazon EC2 service with its characteristics.

2079 Short QNs.

31. What is cloud technology?
32. Mention some of the popularly used cloud computing services.
33. Write short note on host security.
34. Explain difference between traditional corporate data center and cloud data center.
35. What are the major differences between parallel and distributed computing?
36. Write short note on Monitoring-as-a-Service (MaaS).
37. What are different factors involved in disaster recovery planning.
38. What is distributed computing in grid.

2080 very Short QNs.

39. What are the security issues which should be discussed with a cloud-computing vendor?
40. Explain disaster recovery process in cloud computing.
41. What are the key characteristics of PaaS?
42. Discuss the principal characteristics of SOA.
43. What are the SaaS implementation issues?
44. Explain the evolution of cloud computing.

2080
Short QNs.

45. What are the main features of Cloud Computing?
46. What is Elastic Compute Cloud (EC2)?
47. How does the monitoring agent monitor cloud usage?
48. How does cloud-native foundation define Cloud-native applications?
49. What are major differences between traditional corporate data center and cloud data center?
50. How resource replication takes place in Cloud Computing?
51. What is the Platform-as-a-Service?
52. What is the difference between scalability and elasticity?

2081 very
Short QNs.

53. In what manner does parallel and distributed computing differ?
54. What are the implementation issues on SaaS? Explain.
55. Explain the data center virtualization.
56. Explain the disaster recovery planning of cloud computing.
57. Describe possible services that can be achieved through Infrastructure-as-a-Service (IaaS).
58. What do you mean by Elastic IP Addressing? Describe how Elastic IPs work in cloud services.

2081
Short QNs.

Long Questions:

1. What type of development models can be adopted in cloud computing? Describe each of them with suitable examples.
2. What do you mean by an intrusion in a cloud network? How are intrusions in a cloud network detected?
3. Describe the Jericho Cloud Cube Model. What are the advantages of Communication-as-a-Service (CaaS) model?

2076 Long QNs.

4. What is cloud computing? Explain the Platform-as-a-Service (PaaS) in detail.
5. What are different challenges in cloud security and explain cloud security control factors.
6. Explain Jericho Cloud Cube Model.

2077 Long QNs.

7. Explain the term cloud computing with its characteristics and explain difference between grid computing and cloud computing.
8. What is data center? Explain in detail with its building blocks.
9. Explain cloud security challenges and explain why we need cloud security requirements and also explain what are security issues which one should discuss with a cloud-computing vendor.
10. What are different techniques used in implementing network intrusion in the cloud and explain in short MSP?

2079 Long QNs.

11. What are the benefits of using cloud models? Explain the cloud computing deployment models (Types) and also discuss the ethical issues in cloud computing.
12. Describe the different building blocks required to build a data center.
13. What are different approaches used in cloud security control? Explain the purpose of a network intrusion detection system.
14. Write short notes on:
 - a. Disaster management
 - b. Benefits of data center virtualization
 - c. Elastic IP addressing
 - d. IaaS

2080 Long QNs.

15. Explain the need of virtualization. Explain external network virtualization in different layers.
16. Describe features, challenges and benefits of cloud computing.
17. Explain the building blocks required to build a data center.
18. What is Service-Oriented Architecture (SOA)? Explain the principal characteristics of SOA.

2081 Long QNs.

E-Business & E-Governance Previous Year QNs

(Year – 2073, 2074, 2076, 2079, 2080, 2081)

Short Questions:

1. What is E-governance? Explain the advantages of e-governance.
2. What are the key challenges of implementing E-governance in developing countries?
3. Describe the latest IT policy of Nepal in brief.
4. Explain the various levels of E-governance maturity model.
5. Define data warehouse and data mining. Write down their advantages.
6. What is cyber law? What are the various areas that cyber law covers?
7. Define OLAP. What is the importance of OLAP in E-governance?
8. What is E-readiness? Discuss the legal infrastructure preparedness of E-governance.

2073 Short QNs.

9. Define e-governance. Why is e-governance necessary? Write advantages of e-governance.
10. What are the problems encountered in achieving e-governance in the context of Nepal?
11. Discuss the legal infrastructure preparedness of e-governance.
12. Write the application areas of data warehouse and data mining.
13. Describe the action plan of ICT policy 2072.
14. Explain the dissemination of information technology.
15. What is cyber law? Describe the needs of cyber law.
16. Explain maturity model.

2074 Short QNs.

17. Describe evaluation of e-governance in Nepal. Write e-government master plan.
18. Define data mining. Write advantages and disadvantages of data mining.
19. Explain GIDC and data center.
20. Describe human and technological infrastructural preparedness for e-readiness.
21. What is e-governance? Why is e-governance necessary? Describe digital divide.
22. Explain present global trends in e-governance.
23. What is maturity model in e-governance? Describe briefly.
24. What is cyber-crime? Explain cyber law.

2076 Short QNs.

25. Differentiate between E-government and E-governance.
26. Why is E-governance important?
27. What is Government to Citizen (G2C)?
28. Define G2G interaction with example.
29. What are the main challenges and courses that control E-governance in Nepal?
30. Which country has an efficient E-governance system and how?
31. What is the performance NIC?
32. Define digital divide.

2079 very Short QNs.

33. Write about E-governance in health and education.
34. Describe evolution of E-governance and its scope.
35. Discuss present global trends of growth in E-governance.
36. Explain critical flow model of E-governance with suitable example.
37. What changes are brought in the citizen-governance relationship through digital governance.
38. Discuss E-Readiness briefly.

2079 Short QNs.

39. What social impact does e-governance have on society? 40. List the advantages of E-Governance. 41. What is encryption? Write its types. 42. What is the use of digital signatures? 43. Which is the utilization of electronic technology in the administration? 44. What is the performance of NIC? 45. How can e-governance security difficulties be withdrawn? 46. What is the impact of e-Learning sites on students?	2080 very Short QNs.
47. Explain the relation between ICT and governance. 48. Describe e-government maturity model. 49. Write the challenges of e-Governance with specific focus to Nepal. 50. Discuss the scope and content of e-governance. 51. Define cyber law. Why is it important for Nepal? Explain. 52. What is institutional infrastructural preparedness for e-readiness? Describe.	2080 Short QNs.
53. What are the key objectives of e-governance? 54. Describe the concept of digital divide in e-governance. 55. What is Open Government Data (OGD)? 56. How can e-governance improve public service delivery? 57. Explain government to the business model. 58. How e-governance security difficulties can be withdrawn? 59. What is e-readiness in e-governance? 60. Define data system infrastructure.	2081 very Short QNs.
61. Explain in brief about scope and content of e-governance. 62. Differentiate between e-governance and e-government. Explain the issues in e-governance implementations and how to solve these issues. 63. Explain the components of data warehouse. What are applications of data mining in e-governance in context of Nepal? 64. What are pre-requisites of e-readiness for infrastructure development? 65. How can we achieve good governance through e-governance model? Explain. 66. Short note on: a. IT in judiciary b. Electronic service delivery	2081 Short QNs.

Long Questions:

1. Explain six generic models of digital governance in developing countries with their applications. 2. Explain the evolutionary stages of e-governance that are perceived in any governmental organization. 3. If you are elected as an information technology minister, what are your master plans for implementing e-governance in Nepal? Explain in your own view.	2073 Long QNs.
4. Describe data warehouse and data mining. Write advantages of data warehouse and data mining. 5. Describe the scope, content, and present global trends of growth in e-governance. 6. What is model of digital governance? Explain any three models of digital governance.	2074 Long QNs.
7. What is data warehouse? Write application, advantages and disadvantages of data warehouse. 8. What is ICT Act? Mention objective and action plan of ICT Act. 9. Explain models of digital governance. Describe broadcasting dissemination, critical flow and comparative analysis model.	2076 Long QNs.
10. Describe maturity model briefly. 11. Explain the evolutionary stages in E-governance. 12. Describe data warehouse with its advantages and disadvantages. 13. Describe strategies and policy of ICT Act 2072.	2079 Long QNs.
14. Why do we need e-governance? How can we use e-governance to provide efficient services to citizens? Explain using suitable examples. 15. What is e-governance model? Explain critical flow model and comparative analysis model in detail. 16. Define the terms data warehousing and data mining. Discuss key areas that demand the use of data warehousing and data mining. 17. Give the reason why IT infrastructure is needed to implement e-governance. Explain network and computing infrastructure needed to implement e-governance.	2080 Long QNs.
18. Why do we need IT infrastructure to implement E-Governance? Explain network and computing infrastructure needed to implement E-governance. 19. What is E-Governance model? Explain broadcasting/wider dissemination model and critical flow model in detail. 20. Why do we need IT infrastructure to implement E-Governance? Explain network and computing infrastructure needed to implement E-governance 21. Explain human infrastructure preparedness and technical infrastructure preparedness for e-governance.	2080 Long QNs.
